

Polynomials – 3

Symmetric functions, Newton's identities, resultant and discriminant

Homework

8 сентября 2026 года

Theoretical minimum. Elementary symmetric polynomials and the fundamental theorem; Newton's identities relating power sums to the coefficients; Vieta's formulas; the resultant as a polynomial in the coefficients vanishing exactly on a common root; the discriminant and multiple roots; partial fractions $\frac{1}{P} = \sum \frac{1}{P'(x_i)(x-x_i)}$ for simple roots.

1. Prove that $\sum_{i \neq j} \frac{1}{x_i - x_j} = 0$ and $\sum_i \frac{P''(x_i)}{P'(x_i)} = 0$ for a polynomial with simple roots, and express $\sum_i \frac{1}{x_i - a}$ through P and P' .
2. (Mexmat, 2001) Let $k_1, \dots, k_n$ be natural numbers, not necessarily distinct. Prove that the sum of the absolute values of the coefficients of $\prod_{j=1}^n (1 - z^{k_j})$ is at least $2n$.
3. (IMC 2025 shortlist) Let $\alpha_1, \dots, \alpha_n$ be the non-zero distinct roots of a degree n polynomial p with complex coefficients. Prove that $1 = \frac{p(1)}{p(0)} + \sum_{i=1}^n \frac{p(\alpha_i + 1)}{\alpha_i p'(\alpha_i)}$.
4. (Proposed problems) Let $A(z) = z^n + \sum_{k=0}^{n-1} a_k z^k$ have n distinct zeros $z_1, \dots, z_n$ and let $\alpha \neq 0$ be such that no ratio of two zeros equals α . Prove that $\sum_{k=1}^n \left(\frac{1}{A(\alpha z_k) A'(z_k)} + \frac{1}{\alpha A(z_k/\alpha) A'(z_k)} \right) = 0$.
5. (VJIMC 1994) Prove that the roots x_1, x_2 of $x^2 - px - \frac{1}{2}p^2$, where $p \neq 0$ is real, satisfy $x_1^4 + x_2^4 \geq 2 + \sqrt{2}$.
6. (Proposed problems) Let $e_1, \dots, e_n$ be the elementary symmetric polynomials in $X_1, \dots, X_n$ and let M be the ideal they generate in $\mathbb{R}[X_1, \dots, X_n]$. Prove that every monomial of degree strictly greater than $\binom{n}{2}$ belongs to M , and that some monomial of degree $\binom{n}{2}$ does not.

7. (Proposed problems) Let K be a field and let $f \in K[X]$ of degree n have distinct roots $x_1, \dots, x_n$. For $p \leq n$ let $S_1, \dots, S_p$ be the elementary symmetric polynomials in $x_1, \dots, x_p$. Show that $[K(S_1, \dots, S_p) : K] \leq \binom{n}{p}$.
8. Prove that the discriminant of a real cubic is positive exactly when all three roots are real and distinct, negative when one root is real, and explain what this means for Cardano's formula.
9. Let f and g be monic of degrees n and m . Prove that $\text{Res}(f, g)$ equals the determinant of the Sylvester matrix, and deduce that two integer polynomials with a common complex root have resultant zero as an integer.
10. Prove that $\prod_{i < j} (x_i - x_j)^2$ is a symmetric polynomial and express it through the coefficients for $n = 3$; then prove that $\sqrt{D}$ is fixed exactly by the even permutations, so it is rational precisely when the Galois group of f lies in the alternating group.